

DIVERSITY IN LIVING ORGANISMS

Class 9 | Science | Chapter 7

CLASSIFICATION

Classification is the process of arranging organisms into groups based on similarities and differences.

Purpose: makes study of the huge diversity of organisms easier and shows evolutionary relationships.

BASIS OF CLASSIFICATION

- Cell structure: presence/absence of a nuclear membrane (prokaryote/eukaryote).
- Body organisation: unicellular or multicellular.
- Mode of nutrition: autotrophic (self-nourishing) or heterotrophic.
- Body design & level of organisation increase in complexity as we go from simple to advanced organisms.

HIERARCHY OF CLASSIFICATION

Kingdom -> Phylum (Division for plants) -> Class -> Order -> Family -> Genus -> Species

Species is the basic/smallest unit of classification; Kingdom is the broadest.

FIVE KINGDOM CLASSIFICATION

Whittaker's System

FIVE KINGDOMS (R.H. WHITTAKER, 1969)

Kingdom	Key Features	Examples
Monera	Prokaryotic, unicellular, no nuclear membrane	Bacteria, blue-green algae (Cyanobacteria)
Protista	Eukaryotic, unicellular	Amoeba, Paramecium, Diatoms
Fungi	Eukaryotic, multicellular (mostly), heterotrophic; cell wall of chitin	Yeast, mushroom, mould
Plantae	Eukaryotic, multicellular, autotrophic (photosynthesis), cell wall of cellulose	Trees, herbs, shrubs
Animalia	Eukaryotic, multicellular, heterotrophic, no cell wall	Sponges to mammals

KINGDOM PLANTAE -- SUB-GROUPS

Group	Feature
Thallophyta (Algae)	Body not differentiated into root/stem/leaf; e.g. Spirogyra
Bryophyta	'Amphibians of plant kingdom'; e.g. Moss, Marchantia
Pteridophyta	Body differentiated; no seeds; e.g. Fern, Horsetail
Gymnosperms	Naked seeds (not enclosed in fruit); e.g. Pine, Deodar
Angiosperms	Seeds enclosed in fruit; flowering plants; divided into Monocots & Dicots

KINGDOM ANIMALIA

Classification of Animals

MAJOR ANIMAL PHYLA

Phylum	Key Feature	Example
Porifera	Pore-bearing, sessile, aquatic	Sponges
Coelenterata	Hollow body cavity, tissue level of organisation	Hydra, Jellyfish
Platyhelminthes	Flatworms, bilateral symmetry, no true body cavity	Tapeworm, Planaria
Nematoda	Round worms, cylindrical body	Ascaris (roundworm)
Annelida	Segmented body, true body cavity	Earthworm, Leech
Arthropoda	Jointed legs, largest animal phylum	Insects, spiders, crabs
Mollusca	Soft body, often with a shell	Snail, Octopus
Echinodermata	Spiny skin, radial symmetry (adult)	Starfish, Sea urchin
Chordata	Notochord present at some stage	Fish to mammals (incl. humans)

PHYLUM CHORDATA -- SUB-PHYLUM VERTEBRATA

Class	Feature
Pisces (Fish)	Aquatic, gills, cold-blooded, scales
Amphibia	Live in water & land, moist skin, cold-blooded
Reptilia	Dry scaly skin, cold-blooded, lay eggs
Aves (Birds)	Feathers, warm-blooded, forelimbs as wings
Mammalia	Mammary glands, warm-blooded, hair on skin

SUMMARY & REVISION

Key Terms + Quick Recap

CHAPTER SUMMARY -- KEY TERMS

- **Classification** -> grouping organisms by similarities/differences
- **Nomenclature** -> scientific naming (binomial system, by Carl Linnaeus)
- **Species** -> basic unit of classification
- **Five kingdoms** -> Monera, Protista, Fungi, Plantae, Animalia
- **Vertebrata** -> Pisces, Amphibia, Reptilia, Aves, Mammalia

BINOMIAL NOMENCLATURE

Each organism is given a two-part scientific name: **Genus + species** (Genus capitalised, species lowercase, both italicised).

Ex: Humans -- Homo sapiens. Mango -- Mangifera indica.

ONE-LINE RECAP FOR REVISION

Hierarchy: Kingdom > Phylum > Class > Order > Family > Genus > Species.

5 Kingdoms: Monera, Protista, Fungi, Plantae, Animalia.

Complexity increases from Monera to Animalia; naming follows binomial system.

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